Lab 1 - Nonparametric methods in survival analysis

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We will work on the dataset described during the lecture: 42 children affected by Acute Lymphoblastic Leukemia (ALL) in first complete remission (1 CR) randomized (pair matched) to treatment arms (placebo and 6-MP) for remission maintenance (Freirerch et al, 1963).

Lets load the data frei.txt into R and call the "survival" package.

```
library(survival)
frei <- read.table('Acute lymphoblastic data.txt', na.strings = ".", header = TRUE, row.names = NULL)
names(frei)
```

```
## [1] "upn"    "treat"  "time"   "status"
```

```
head(frei)
```

```
##   upn treat time status
## 1   1     1    1      1
## 2   2     1    1      1
## 3   3     1    2      1
## 4   4     1    2      1
## 5   5     1    3      1
## 6   6     1    4      1
```

Question 1

Working on the pooled sample, estimate and plot the survival function by the Kaplan-Meier method with Greenwood confidence bands.

```
fit <- survfit(Surv(time, status) ~ 1, data = frei)
summary(fit)
```

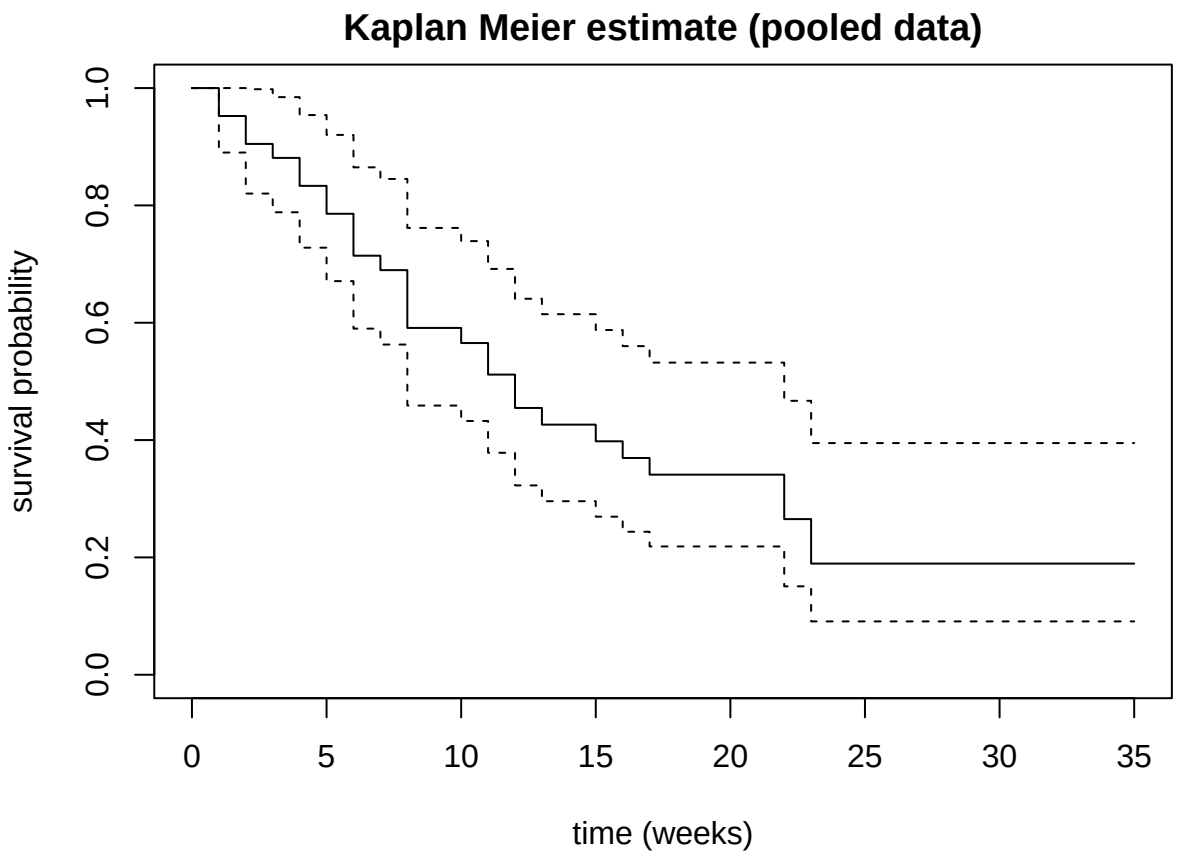
```
## Call: survfit(formula = Surv(time, status) ~ 1, data = frei)
##
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     1     42       2    0.952  0.0329       0.8901        1.000
##     2     40       2    0.905  0.0453       0.8202        0.998
##     3     38       1    0.881  0.0500       0.7883        0.985
##     4     37       2    0.833  0.0575       0.7279        0.954
##     5     35       2    0.786  0.0633       0.6709        0.920
##     6     33       3    0.714  0.0697       0.5899        0.865
##     7     29       1    0.690  0.0715       0.5628        0.845
##     8     28       4    0.591  0.0764       0.4588        0.762
##    10     23       1    0.565  0.0773       0.4325        0.739
##    11     21       2    0.512  0.0788       0.3783        0.692
##    12     18       2    0.455  0.0796       0.3227        0.641
##    13     16       1    0.426  0.0795       0.2958        0.615
```

```
##    15     15       1    0.398  0.0791       0.2694        0.588
##    16     14       1    0.369  0.0784       0.2437        0.560
##    17     13       1    0.341  0.0774       0.2186        0.532
##    22      9       2    0.265  0.0765       0.1507        0.467
##    23      7       2    0.189  0.0710       0.0909        0.395
```

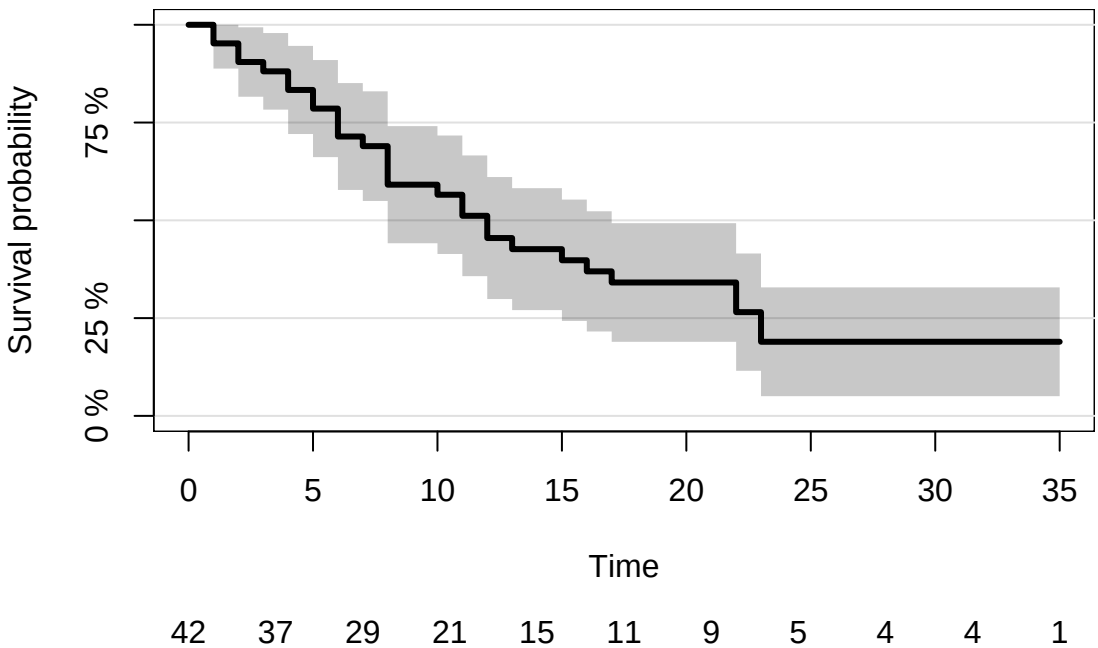
```r
names(fit)
```

```
##  [1] "n"         "time"      "n.risk"    "n.event"   "n.censor"  "surv"
##  [7] "std.err"   "cumhaz"    "std.chaz"  "type"      "logse"     "conf.int"
## [13] "conf.type" "lower"     "upper"     "t0"        "call"
```

```r
par(mar = c(4, 4, 2, 2))
plot(fit, xlab = 'time (weeks)', ylab = 'survival probability')
title('Kaplan Meier estimate (pooled data)')
```




```r
library(prodlim)
fit.pl <- prodlim(Hist(time, status) ~ 1, data = frei)
plot(fit.pl)
```



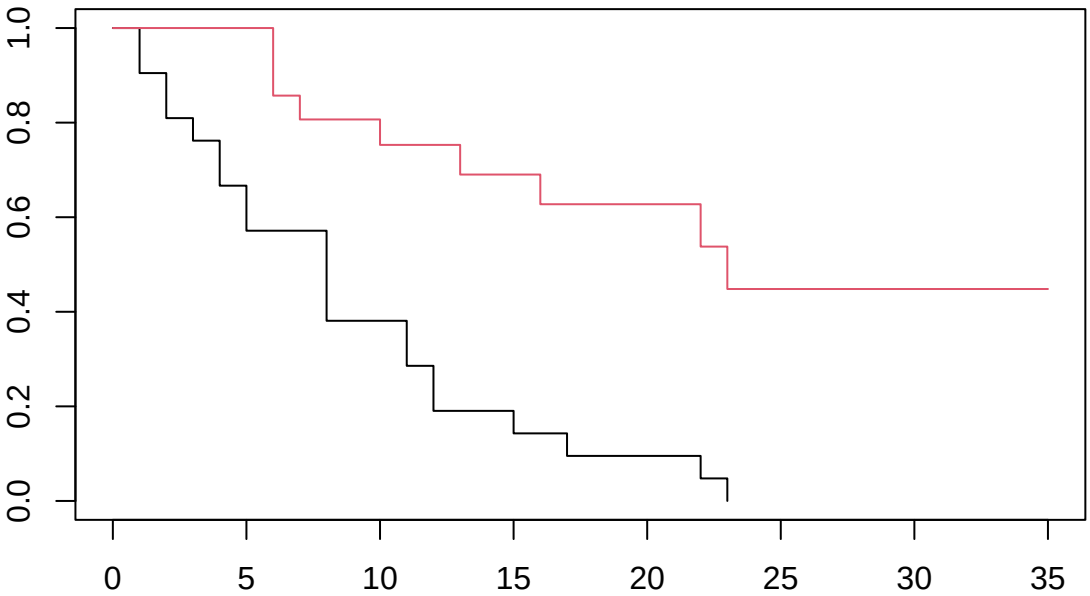
Question 2

For each treatment, estimate and plot the survival function by the KM method.

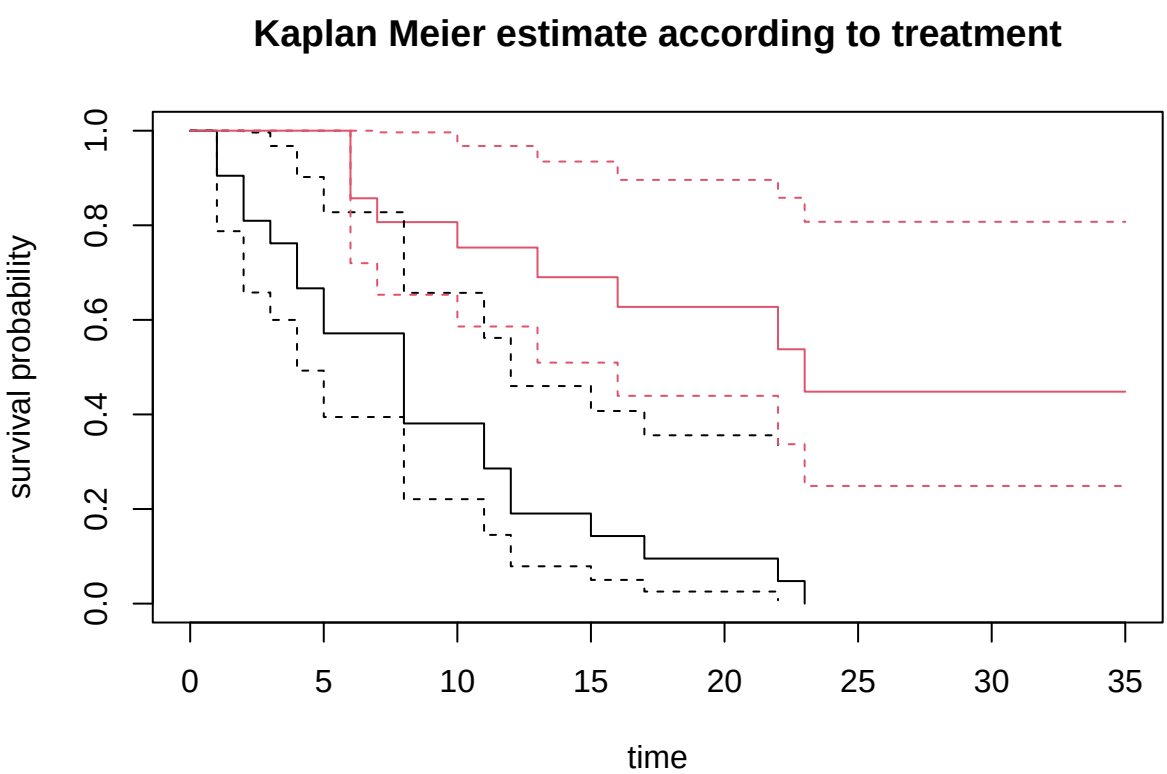
```
fit <- survfit(Surv(time, status) ~ treat, data = frei)
summary(fit)
```

```
## Call: survfit(formula = Surv(time, status) ~ treat, data = frei)
##
##                 treat=1
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     1     21       2   0.9048  0.0641      0.78754        1.000
##     2     19       2   0.8095  0.0857      0.65785        0.996
##     3     17       1   0.7619  0.0929      0.59988        0.968
##     4     16       2   0.6667  0.1029      0.49268        0.902
##     5     14       2   0.5714  0.1080      0.39455        0.828
##     8     12       4   0.3810  0.1060      0.22085        0.657
##    11      8       2   0.2857  0.0986      0.14529        0.562
##    12      6       2   0.1905  0.0857      0.07887        0.460
##    15      4       1   0.1429  0.0764      0.05011        0.407
##    17      3       1   0.0952  0.0641      0.02549        0.356
##    22      2       1   0.0476  0.0465      0.00703        0.322
##    23      1       1   0.0000     NaN           NA           NA
##
##                 treat=2
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     6     21       3    0.857  0.0764        0.720        1.000
##     7     17       1    0.807  0.0869        0.653        0.996
##    10     15       1    0.753  0.0963        0.586        0.968
##    13     12       1    0.690  0.1068        0.510        0.935
##    16     11       1    0.627  0.1141        0.439        0.896
##    22      7       1    0.538  0.1282        0.337        0.858
##    23      6       1    0.448  0.1346        0.249        0.807
```

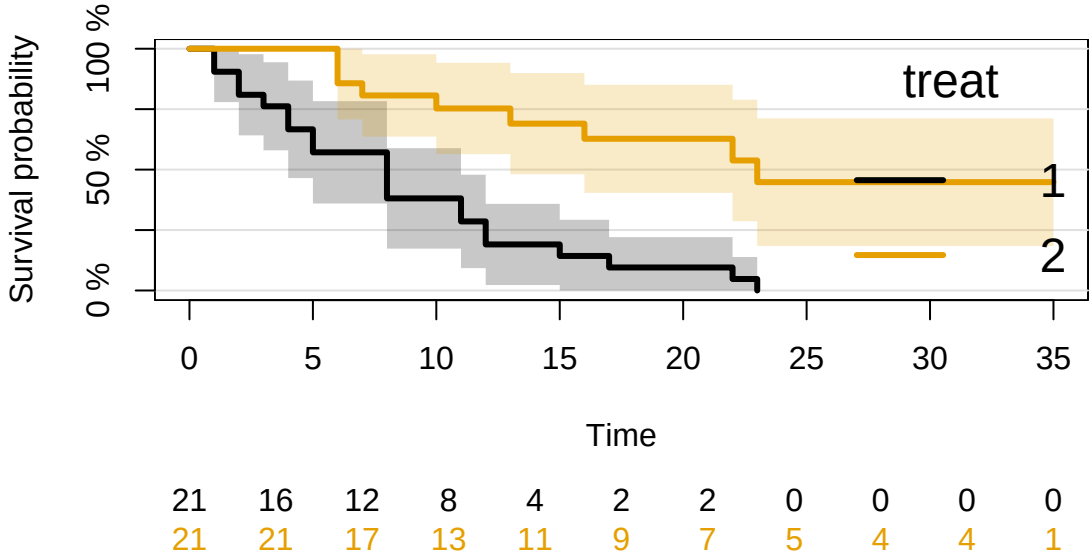
```r
plot(fit, col = c(1, 2))
```




```r
plot(c(0, fit[treat=1]$time), c(1, fit[treat=1]$surv), type = 's', ylim = c(0, 1),
     xlim = c(0, 35), xlab = 'time', ylab = 'survival probability', col = 1)
lines(c(0, fit[treat=2]$time), c(1, fit[treat=2]$surv), type = 's', col = 2)
lines(c(0, fit[treat=1]$time), c(1, fit[treat=1]$lower), type = 's', lty = 2, col = 1)
lines(c(0, fit[treat=1]$time), c(1, fit[treat=1]$upper), type = 's', lty = 2, col = 1)
lines(c(0, fit[treat=2]$time), c(1, fit[treat=2]$lower), type = 's', lty = 2, col = 2)
lines(c(0, fit[treat=2]$time), c(1, fit[treat=2]$upper), type = 's', lty = 2, col = 2)
title('Kaplan Meier estimate according to treatment')
```



```r
library(prodlim)
fit.pl <- prodlim(Hist(time, status) ~ treat, data = frei)
plot(fit.pl)
```



Question 3

Contrast in a plot the number of censored subjects and those at risk in the two groups.

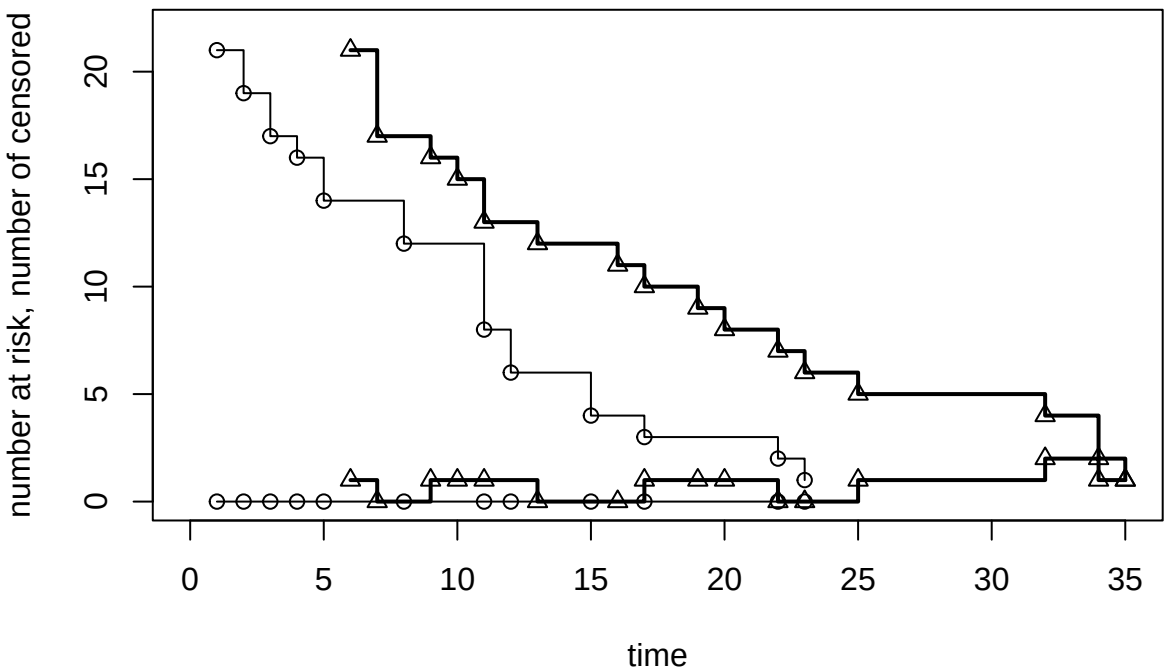
```r
# At risk
plot(fit[treat=1]$time, fit[treat=1]$n.risk, xlab = 'time',
     ylab = 'number at risk, number of censored',
     type = 's', ylim = c(0, 22), xlim = c(0, 35))
points(fit[treat=1]$time, fit[treat=1]$n.risk)

lines(fit[treat=2]$time, fit[treat=2]$n.risk, type = 's', lwd = 2)
points(fit[treat=2]$time, fit[treat=2]$n.risk, pch = 2)

# Censored subjects (reverse estimation)
fit.rev <- survfit(Surv(time, status == 0) ~ treat, data = frei)

lines(fit.rev[treat=1]$time, fit.rev[treat=1]$n.event, type = 's')
points(fit.rev[treat=1]$time, fit.rev[treat=1]$n.event)

lines(fit.rev[treat=2]$time, fit.rev[treat=2]$n.event, type = 's', lwd = 2)
points(fit.rev[treat=2]$time, fit.rev[treat=2]$n.event, pch = 2)
```



Question 4

Working on the pooled sample, estimate and plot the cumulative hazard function: a. by the Nelson-Aalen estimate b. by $-\log(S(t))$

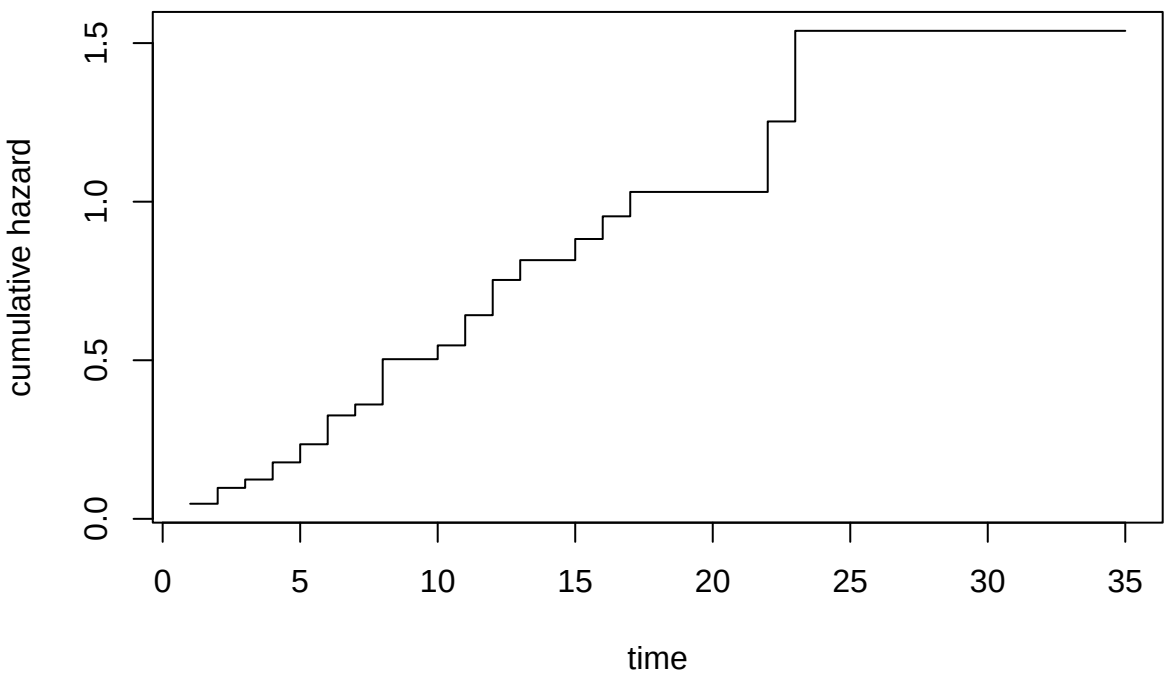
```
fit.pool <- survfit(Surv(time, status) ~ 1, data = frei)

# Nelson-Aalen
na.haz <- cumsum(fit.pool$n.event / fit.pool$n.risk)
cbind(fit.pool$time, na.haz)
```

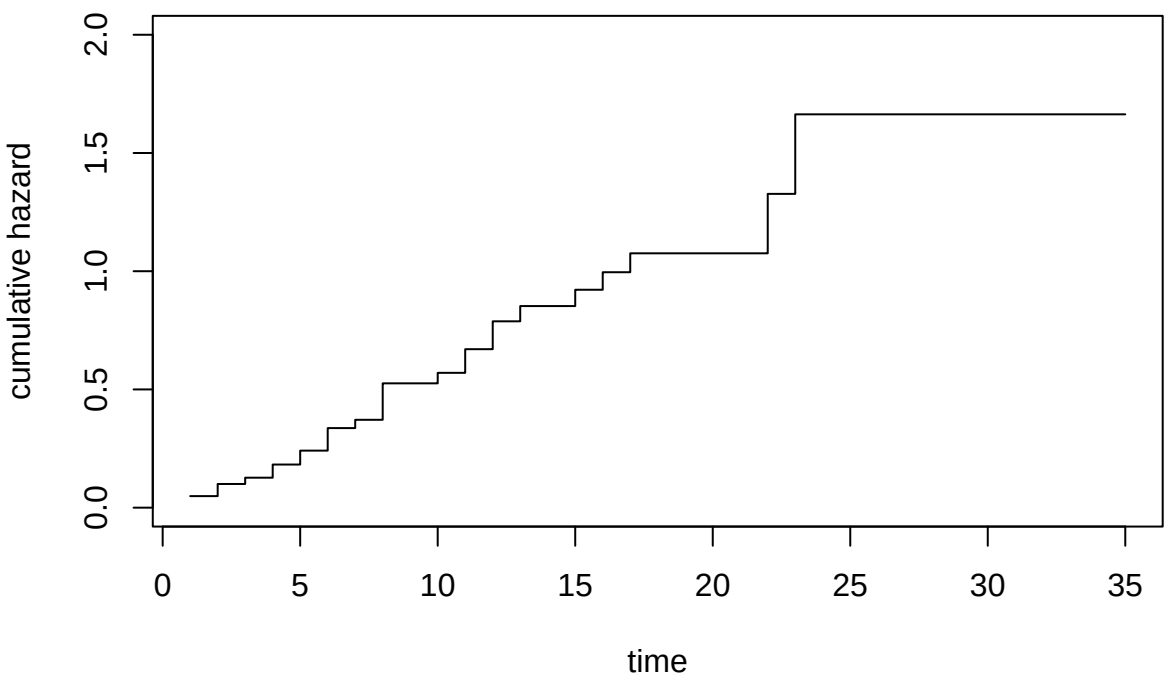
```
##              na.haz
##  [1,]  1 0.04761905
##  [2,]  2 0.09761905
##  [3,]  3 0.12393484
##  [4,]  4 0.17798889
##  [5,]  5 0.23513175
##  [6,]  6 0.32604084
##  [7,]  7 0.36052360
##  [8,]  8 0.50338074
##  [9,]  9 0.50338074
## [10,] 10 0.54685900
## [11,] 11 0.64209710
## [12,] 12 0.75320821
## [13,] 13 0.81570821
## [14,] 15 0.88237487
## [15,] 16 0.95380345
## [16,] 17 1.03072652
## [17,] 19 1.03072652
## [18,] 20 1.03072652
## [19,] 22 1.25294875
## [20,] 23 1.53866303
## [21,] 25 1.53866303
```

```
## [22,] 32 1.53866303
## [23,] 34 1.53866303
## [24,] 35 1.53866303
```

```r
plot(fit.pool$time, na.haz, type = 's', xlab = 'time', ylab = 'cumulative hazard')
```

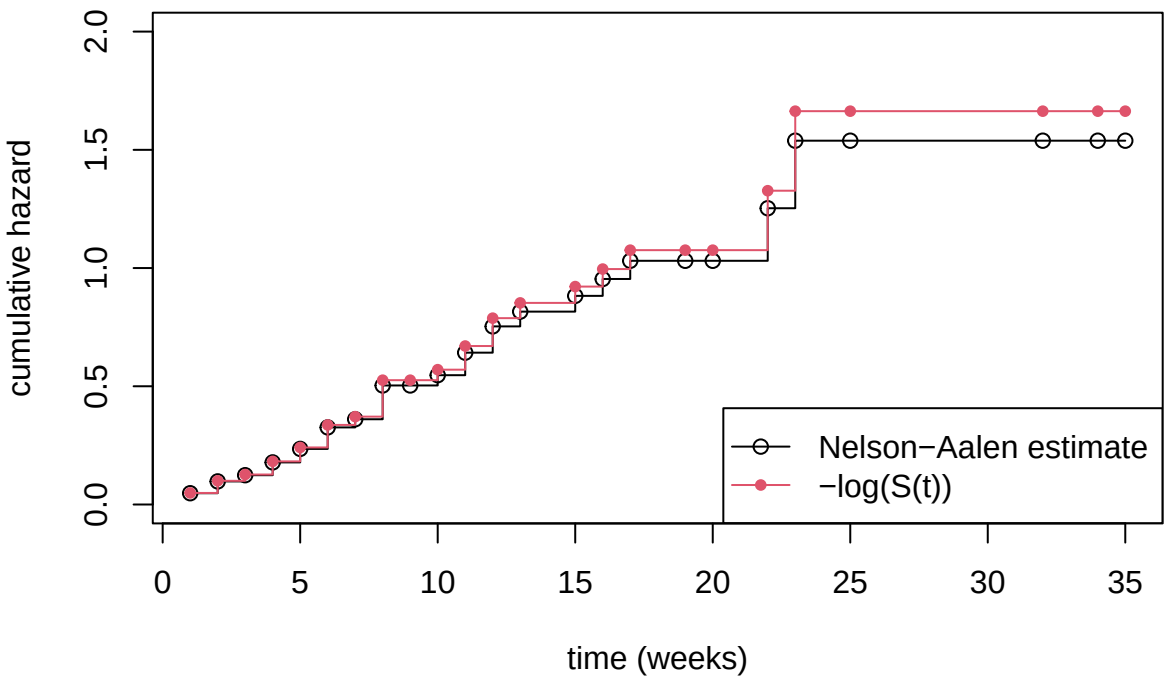



```r
# -log(S(t))
cumhaz <- -log(fit.pool$surv)
plot(fit.pool$time, cumhaz, type = 's', ylim = c(0, 2), xlab = 'time', ylab = 'cumulative hazard')
```




```r
plot(fit.pool$time, na.haz, type = 's', xlab = 'time (weeks)',
     ylab = 'cumulative hazard', ylim = c(0, 2))
```

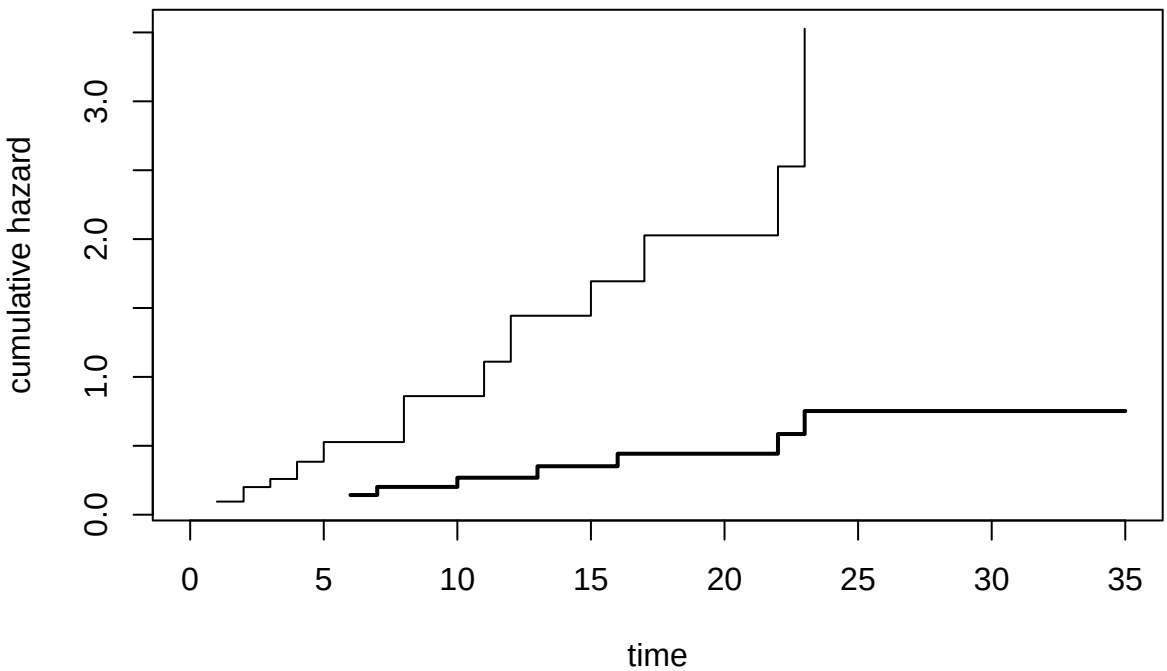
```r
points(fit.pool$time, na.haz)
lines(fit.pool$time, cumhaz, type = 's', col = 2)
points(fit.pool$time, cumhaz, pch = 20, col = 2)
legend("bottomright",
       legend = c("Nelson-Aalen estimate", "-log(S(t))"),
       col = c(1, 2), lty = 1, pch = c(1, 20))
```



Question 5

Estimate and plot the cumulative hazard in the two treatment groups.

```r
na.haz1 <- cumsum(fit[treat=1]$n.event / fit[treat=1]$n.risk)
na.haz2 <- cumsum(fit[treat=2]$n.event / fit[treat=2]$n.risk)
plot(fit[treat=1]$time, na.haz1, xlab = 'time', ylab = 'cumulative hazard',
     type = 's', xlim = c(0, 35))
lines(fit[treat=2]$time, na.haz2, type = 's', lwd = 2)
```

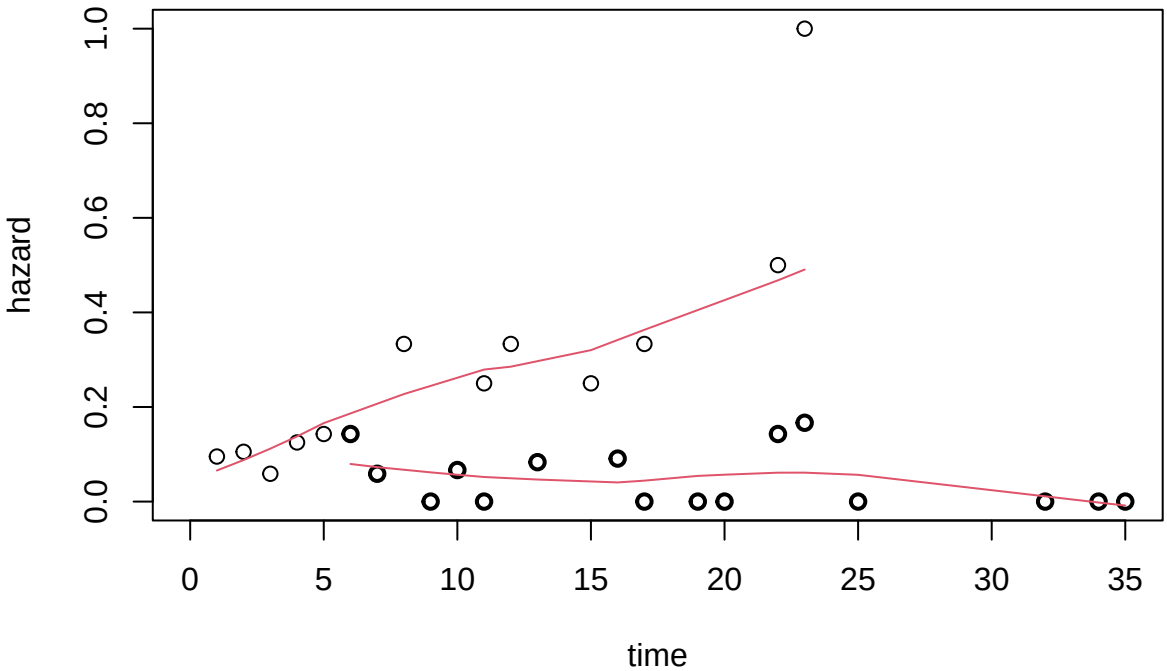


Question 6

Provide a smoothed estimate of the hazard function of the two treatment groups.

```
haz1 <- fit[treat=1]$n.event / fit[treat=1]$n.risk
haz2 <- fit[treat=2]$n.event / fit[treat=2]$n.risk

plot(fit[treat=1]$time, haz1, xlab = 'time', ylab = 'hazard',
     xlim = c(0, 35), ylim = c(0, 1))
lines(lowess(cbind(fit[treat=1]$time, haz1)), col = 2)
points(fit[treat=2]$time, haz2, lwd = 2)
lines(lowess(cbind(fit[treat=2]$time, haz2)), col = 2)
```



Question 7

Estimate the failure rate under the hypothesis of constant hazard and interpret in terms of events/person weeks.

```
# Placebo (treat = 1)
tot.time.at.risk <- sum(frei$time[frei$treat == 1])
tot.events       <- sum(frei$status[frei$treat == 1])
f.rate           <- tot.events / tot.time.at.risk
cbind(tot.events, tot.time.at.risk, f.rate)
```

```
##      tot.events tot.time.at.risk    f.rate
## [1,]         21              182 0.1153846
```

```
# 6-MP (treat = 2)
tot.time.at.risk <- sum(frei$time[frei$treat == 2])
tot.events       <- sum(frei$status[frei$treat == 2])
f.rate           <- tot.events / tot.time.at.risk
cbind(tot.events, tot.time.at.risk, f.rate)
```

```
##      tot.events tot.time.at.risk     f.rate
## [1,]          9              359 0.02506964
```

Question 8

Test the null hypothesis of absence of treatment effect (6-MP vs PL) using a nonparametric hypothesis test (log-rank test).

```
survdiff(Surv(time, status) ~ treat, data = frei)
```

```
## Call:
## survdiff(formula = Surv(time, status) ~ treat, data = frei)
##
##          N Observed Expected (O-E)^2/E (O-E)^2/V
## treat=1 21       21     10.7      9.77      16.8
```

```
## treat=2 21        9     19.3      5.46      16.8
##
##  Chisq= 16.8  on 1 degrees of freedom, p= 4e-05
```